

VAE (Maths)

[Kingma & Welling]

In Plain Autoencoder the decoder is deterministic: one z in, one $\hat{x}$ out.

In Generative view we make it "Probabilistic": given a latent z , the decoder defines a whole distribution over data,

$$p_{\theta}(x|z)$$

θ = decoder network's weights

$$1. \text{ draw } \rightarrow z \sim N(0, I)$$

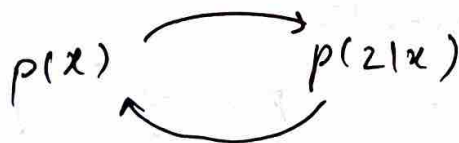
$$2. \text{ draw } \rightarrow x \sim p_{\theta}(x|z)$$

Marginal Likelihood (Evidence)

$$p_{\theta}(x) = \int p_{\theta}(x|z)p(z)dz$$

Intractable issue (Chicken-Egg Problem)

$$p_{\theta}(z|x) = \frac{p_{\theta}(x|z)p(z)}{p_{\theta}(x)}$$



Variational Inference & Amortization $\rightarrow$

App. Posterior $\rightarrow$

$$q_{\phi}(z|x) = N(z; \mu_{\phi}(x), \sigma^2_{\phi}(x))$$

Mean Variance

We tune ϕ so, $q_{\phi}(z|x)$ hugs the true posterior $p_{\theta}(z|x)$

Closeness is measured by KL $\rightarrow$

$$\text{Min-}\phi \quad KL(q_{\phi}(z|x) || p_{\theta}(z|x))$$

Amortized Inference $\rightarrow$

Classical Variational Inference Each individual z , you can run a fresh optimization to find the best μ, σ for that one point.
With 60,000 images 60,000 separate optimizations.

VAE Trick - take a single Network (Encoder) ^{Instantly}
that takes any $x \rightarrow$ outputs $\mu - \phi(x), \sigma - \phi(x)$.

"Amortized" just means that one-time-cost spread across - all datapoints arrangement.

So it's an,

Amortized - Variational Inference Machine

ELBO $\rightarrow$ (Evidence Lower Bound)

Jensen's Inequality

$$\log p(x) = \log \int p(x, z) dz.$$

$$= \log \int q(z|x) \cdot \left[\frac{p(x, z)}{q(z|x)} \right] dz. \quad \text{Multiple } q(z|x).$$

$\hat{p}(x)$ $\hat{p}(x)$
Log of an Expectⁿ $\rightarrow$

$$\Rightarrow \log \mathbb{E}_{q(z|x)} \left[\frac{p(x, z)}{q(z|x)} \right] \geq \mathbb{E}_{q(z|x)} \left[\log \left(\frac{p(x, z)}{q(z|x)} \right) \right]$$

(Jensen's Inequality)

= ELBO.

$\therefore$ 'Log' function is concave, and for any concave function the log of an avg $\geq$ avg of the logs. (

Ex $\rightarrow \log$ of the Avg $\rightarrow \log\left(\frac{1+100}{2}\right) = \log(50.5) = 1.70$
 Avg. of the logs $\rightarrow (\log 1 + \log 100)/2 = (0+2)/2 = 1$.
 $\log \mathbb{E}[Y] = 1.70 \geq 1.00 = \mathbb{E}[\log Y]$

Jensen's Inequality Probabilistic form $\rightarrow$
 $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$.

So,
 $\log p_0(x) \geq \mathbb{E}_{q(z|x)} \left[\log \left(\frac{p_0(x, z)}{q_\phi(z|x)} \right) \right]$
 $\mathbb{E}_{q(z|x)} \left[\log \left(\frac{p_0(x|z) p(z)}{q_\phi(z|x)} \right) \right]$

$= \mathbb{E}_{q(z|x)} \left[\log(p_0(x|z)) \right]$
 $+ \mathbb{E}_{q(z|x)} \left[\log \frac{p(z)}{q(z|x)} \right]$.
 Flip $\rightarrow$

$\Rightarrow \mathbb{E}_{q(z|x)} [\log p_0(x|z)]$
 $- \text{DKL}(q(z|x) || p(z))$.

$\uparrow$
 Maximizing
 the Reconst.
 of the Decoder

$\uparrow$
 Minimizing the
 Distance b/w the
 learned Distance & the
 prior belief we have over
 the latent variable.

Importance Sampling $\rightarrow$

You want an $\mathbb{E}$ under a target distribⁿ

$$p - \mathbb{E}_{z \sim p} [f(z)] = \int f(z) p(z) dz$$

But sampling from 'p' is bad (inefficient, like the prior) or impossible.

$$\mathbb{E}_{z \sim p} [f(z)] = \int f(z) p(z) dz$$

$$= \int f(z) \left[\frac{p(z)}{q(z)} \right] \cdot \underbrace{q(z) dz}_{p(z)}$$

$$= \mathbb{E}_{z \sim q} \left[f(z) \cdot \frac{p(z)}{q(z)} \right]$$

$\propto$ Importance Ratio.

So you draw from q & Avg $f(z)$ times the importance Weight

$$w(z) = \frac{p(z)}{q(z)}$$

Importance
ELBO & Sampling $\rightarrow$

ELBO is nothing but the log of an importance sampling estimator of $p(x)$ - the ratio sitting in your handwritten derivation was an importance weight all along.

Reparameterization Trick

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim N(0, I)$$

Closed Form KL

$$KL(\underbrace{N(\mu, \sigma^2)}_{q=N(\mu, \sigma^2)} \parallel \underbrace{N(0, 1)}_{\text{Prior } N(0, 1)}) = \frac{1}{2} (\mu^2 + \sigma^2 - \log \sigma^2 - 1)$$

Expectⁿ → Weighted Avg of a function's values, where weights are probabilities

$$E[X] = \sum_x x \cdot p(x)$$

Continuous Case.

$$E_{p(x)}[f(x)] = \int f(x) p(x) dx$$

We Appx. with samples - That integral is usually impossible to compute analytically.

So we use Monte Carlo →

$$E_{p(x)}[f(x)] \approx \frac{1}{N} \sum_{i=1}^N f(x_i), x_i \sim p(x)$$

Expectation = Weighted Avg.

Monte Carlo = Appxⁿ by watching/sampling.

Prior Sampling →

Detective Analogy.

Prior Sampling is like throwing darts blindly. Most darts miss! In high dimensions, almost all darts miss.

Ex →

You Run a Coffee Shop.
Every Customer orders a drink that costs \$3, \$5, & \$8.

50% order that \$3

30% order that \$5

20% order that \$8

Weighted Avg.

$$0.5 \times 3 + 0.3 \times 5 + 0.2 \times 8 = 4.6$$

Now you don't know prob.
You just stand at the counter & watch 1000 Customers -

$$E[\text{Spend}] \approx \frac{1}{1000} \sum_{i=1}^{1000} \text{cost}_i$$

(Monte Carlo Estimation)

When you can't compute an Expectatiⁿ exactly, you sample from the distribution & Avg.
Sampling = Watching Real Customers.

$$p(x) = \int p(x|z) p(z) dz = E_{p(z)}[p(x|z)]$$

App. with prior sample -

$$p(x) \approx \frac{1}{K} \sum_{i=1}^K p(x|z_i), z_i \sim p(z)$$

Importance Sampling $\rightarrow$

Detective Analogy.

Suspect is somewhere someone who
lives within 5 miles.

Now Sampling from Biased Distⁿ.

If the jist counts suspects in this
5 mile radius, she'll overcount
locals & miss out-of-towners.

Local who is common $\rightarrow$

Moderate/Low Weight.

Rare tourist (Surprise $\uparrow$) $\rightarrow$

High Weight.

Amortized Inference.

We want $p(z|x)$

Be Bayes

$$p(z|x) = \frac{p(x|z) p(z)}{p(x)} \leftarrow \text{Intractable}$$

Classical Solⁿ - Variational Inference

App $p(z|x)$ to $q(z|x)$:

$$q(z|x) = \mathcal{N}(\mu_x, \sigma_x^2)$$

So for each x we need to compute

μ_x & σ_x to make $q(z|x) \approx p(z|x)$

Amortized Trick

Train a NN that maps $x \rightarrow (\mu_x, \sigma_x)$

$$\mu_x, \log \sigma_x^2 = \text{Encoder NN}(x)$$

$$E_{p(x)}[f(x)]$$

$$= E_{q(x)} \left[f(x) \cdot \frac{p(x)}{q(x)} \right]$$

$\uparrow$
Imp. weight

MC Version

$$E_{p(x)}[f(x)] \approx \frac{1}{N} \sum_{i=1}^N f(x_i) \cdot \frac{p(x_i)}{q(x_i)} \quad x_i \sim q(x)$$

$\mathcal{D}$ = Sample from easy q ,
reweight by p/q to correct the
bias.